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A lightweight privacy-preserving reranker enables customizable hybrid retrieval for enterprise document search

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Abstract

Enterprise document retrieval systems face a critical challenge: how to deliver high-quality search results while maintaining strict privacy requirements such as regulatory compliance and complete data sovereignty. Existing solutions either rely on cloud-based APIs that violate privacy constraints or deploy large-scale models (278 M+ parameters) that are impractical for on-premise deployment. We introduce HR+QDA (Hybrid Retrieval with Query-Document Attention), a privacy-preserving retrieval system that enables organizations to deploy customizable document search while maintaining complete data control. HR+QDA combines BM25 sparse retrieval with a lightweight 1.77 M-parameter cross-attention reranker that can be trained on-premise in 13 min with as few as 1,500 labeled examples. Evaluated on NFCorpus nutrition domain (2,914 queries), HR+QDA achieves 91% of state-of-the-art performance (MRR 0.744 vs. 0.814) with 157× fewer parameters than BGE-reranker-base, 2.3× faster inference (29ms vs. 68ms), and crucially, the ability to customize on private domain-specific data without external transmission. Our results demonstrate that lightweight, privacy-preserving customization provides a practical path for enterprise retrieval deployment across domains including healthcare, finance, legal, and technical documentation that addresses the critical gap between cloud-based accuracy and on-premise privacy requirements.

Keywords Domain-specific information retrieval, Privacy-preserving ML, Enterprise AI, Regulatory compliance, Domain adaptation, Lightweight neural reranking

1 Introduction

Organizations across industries increasingly require sophisticated document retrieval systems to navigate vast repositories of domain-specific literature, technical documentation, and proprietary knowledge bases. The demand for information retrieval (IR) systems that can deliver precise, contextually relevant answers from large-scale corpora has accelerated across knowledge-intensive domains including healthcare, finance, legal services, and technical research [1, 2]. However, deploying such systems faces a fundamental tension between performance and privacy. State-of-the-art neural retrieval models achieve impressive accuracy but typically operate through cloud APIs (OpenAI, Cohere,



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Anthropic) that require transmitting sensitive data to external servers a clear violation of regulatory compliance policies (GDPR, HIPAA, financial regulations) and institutional data sovereignty requirements [3, 4].

The modern IR landscape has largely evolved beyond single-paradigm systems, converging on a consensus that hybrid approaches combining sparse lexical matching with dense semantic retrieval represent the state-of-the-art [5–7]. This convergence reflects fundamental complementarity: sparse methods like BM25 excel at exact term matching and rare entity retrieval, while dense neural encoders capture semantic similarity and query reformulations [8, 9]. Recent benchmarks demonstrate that hybrid systems consistently outperform single-method approaches by 5–15% across diverse domains, with learned sparse representations (SPLADE) and late interaction models (ColBERT) achieving particularly strong zero-shot generalization [10–12]. The BEIR heterogeneous benchmark has established hybrid retrieval as essential for robust cross-domain performance, where neither sparse nor dense methods alone maintain consistent effectiveness.

The alternative of deploying large-scale models on-premise presents severe practical challenges. Current state-of-the-art rerankers like BGE-reranker-base contain 278 M parameters, requiring significant computational infrastructure, lengthy training times (5 + hours), and substantial labeled datasets (10 K+ examples) [13, 14]. For most organizations, these requirements are prohibitive, creating a critical gap between theoretical capability and practical deployment. Furthermore, while hybrid retrieval architectures provide superior first-stage candidate generation, the subsequent reranking stage typically relies on massive cross-encoder models that compound deployment challenges through inference latency (60–100ms per query) and resource demands [15–17].

We address this challenge with HR + QDA (Hybrid Retrieval with Query-Document Attention), a privacy-preserving retrieval system designed specifically for enterprise deployment. HR + QDA makes three key contributions. First, it implements a complete privacy-preserving architecture through on-premise deployment using publicly available pre-trained models, enabling regulatory-compliant customization without external data transmission. Second, it achieves lightweight efficiency with a 1.77 M-parameter reranker, 157× smaller than state-of-the-art alternatives that maintains 91% of SOTA performance while requiring only 29ms per query (2.3× faster than BGE-reranker-base). Third, it demonstrates practical scalability through training convergence in 13 min with 1,500 examples, achievable through 2–3 weeks of user feedback collection in operational deployment (Fig. 1).

Evaluated on NFCorpus (2,914 nutrition domain queries), HR + QDA demonstrates that lightweight, customizable reranking provides a practical solution for enterprise retrieval. While generic semantic search baselines achieve higher performance on this public benchmark (MRR 0.846), they represent frozen models that cannot be customized for domain-specific terminology, proprietary knowledge bases, or evolving institutional vocabularies [18]. HR + QDA bridges this critical gap, enabling privacy-preserving customization with competitive accuracy (MRR 0.744), fast inference, and minimal

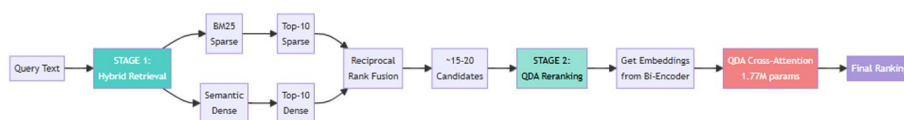


Fig. 1 Two-stage HR+QDA architecture: hybrid retrieval via BM25 and semantic search, followed by QDA reranking

training overhead—precisely the requirements for real-world enterprise deployment across healthcare, finance, legal, and technical domains.

2 Related work

This research is situated at the intersection of four pivotal areas in modern information retrieval: the design of hybrid retrieval architectures, the domain-specific adaptation of dense retrievers, the development of advanced reranking models, and the critical trade-offs involved in model fine-tuning.

2.1 The evolution toward hybrid retrieval architectures

The trajectory of information retrieval over the past decade has been characterized by a clear evolution from single-paradigm systems to sophisticated hybrid architectures that integrate lexical and semantic signals. For decades, the field was dominated by sparse retrieval models, with BM25 serving as the remarkably robust benchmark based on probabilistic relevance theory [19, 20]. BM25 scores documents through term frequency and inverse document frequency calculations, excelling at explicit keyword matching with computational efficiency. Its enduring strength in precise term matching scenarios ensures continued relevance in modern search systems. However, BM25 suffers from the fundamental “vocabulary mismatch” problem and the inability to bridge semantic gaps when queries and relevant documents use different terminology [21].

The advent of large pre-trained language models catalyzed a paradigm shift toward dense retrieval. Dense Passage Retrieval (DPR) and bi-encoder architectures leverage transformer models to map queries and documents into shared semantic vector spaces where relevance is determined by proximity metrics like cosine similarity. This approach effectively addresses vocabulary mismatch by capturing conceptual meaning rather than requiring lexical overlap. However, dense methods introduce their own challenges: high computational requirements for embedding generation, potential loss of critical keyword signals, and significant performance degradation under domain shift [22].

Recognizing the complementary nature of these paradigms, the field has converged on hybrid systems as the state-of-the-art approach [5, 6]. Recent work demonstrates that combining sparse and dense signals yields 5–15% improvements across diverse benchmarks, with techniques like Reciprocal Rank Fusion (RRF) providing effective score combination strategies [23, 24]. The BEIR heterogeneous benchmark has established that neither sparse nor dense methods alone maintain robust cross-domain performance, making hybridization essential [1]. Advanced hybrid architectures include learned sparse representations (SPLADE) that achieve neural learning within inverted index structures, and late-interaction models (ColBERT) that defer query-document matching until after independent encoding. Most recently, unified models like BGE-M3 demonstrate that single architectures can simultaneously produce sparse, dense, and multi-vector representations, while retrieval-augmented generation systems increasingly rely on hybrid retrieval for robust document selection [5].

2.2 Neural information retrieval

While hybrid retrieval generates high-quality candidate sets, reranking with neural cross-encoders remains critical for achieving state-of-the-art accuracy. Cross-encoder architectures jointly encode query-document pairs, enabling fine-grained interaction

modeling that substantially improves ranking precision [25]. BERT-based rerankers pioneered this approach, with models like monoT5 and BGE-reranker-base achieving top performance on standard benchmarks through large-scale pre-training (278 M parameters for BGE). However, these gains come with severe deployment penalties: cross-encoders require separate forward passes for each query-document pair, resulting in 60-100ms inference latency per query and prohibitive memory footprints for enterprise systems [26].

Recent efforts toward efficient reranking have explored multiple directions. Knowledge distillation approaches transfer capabilities from large language model rerankers into smaller cross-encoders, achieving 173× speedups while maintaining competitive accuracy [14]. Listwise reranking methods process multiple candidates jointly, reducing latency through batched inference [27]. Lightweight alternatives like title-based rerankers achieve 20–40× speedups by operating on compressed document representations [22]. Late-interaction rerankers such as ColBERT decouple encoding from matching, enabling pre-computation of document representations and yielding sub-10ms query latency. Despite these advances, most efficient rerankers either sacrifice substantial accuracy (5–10% MRR degradation), require complex serving infrastructure, or remain too large (110 M+ parameters) for resource-constrained enterprise deployment [28].

2.3 Domain-specific information retrieval and adaptation

Enterprise retrieval systems face unique challenges stemming from specialized terminology, evolving vocabularies, and domain-specific relevance patterns that differ markedly from general web search. Medical retrieval systems must navigate clinical terminology and drug nomenclature, legal retrieval requires understanding of case law citations and statutory language, while technical documentation demands precise handling of product codes and internal abbreviations. Standard dense retrievers trained on general corpora exhibit significant performance degradation on specialized domains, often dropping 20–40% in effectiveness metrics when evaluated on domain-specific test sets [1].

Domain adaptation approaches typically follow three paradigms. Full fine-tuning of large pre-trained models yields strong in-domain performance but requires substantial labeled data (10 K+ examples), lengthy training (5+ hours), and risks catastrophic forgetting of general knowledge. Continued pre-training on domain corpora offers partial adaptation but demands even larger unlabeled datasets and computational resources. Parameter-efficient methods like adapters and prompt tuning reduce training costs but still require careful hyperparameter selection and domain-specific validation. Critically, all these approaches assume access to domain-specific training infrastructure and expertise, creating barriers for organizations with limited machine learning resources. Recent work on self-supervised domain adaptation through pseudo-relevance feedback and synthetic query generation [29] shows promise for reducing annotation requirements, yet these methods remain underexplored for privacy-sensitive enterprise environments where external model APIs are prohibited.

2.4 Privacy-preserving machine learning

Privacy concerns in retrieval systems have intensified as organizations face stringent regulatory requirements (GDPR, HIPAA, financial sector regulations) and increasing scrutiny over data sovereignty [2]. Cloud-based retrieval APIs from commercial providers

(OpenAI, Cohere, Anthropic) require transmitting sensitive queries and documents to external servers, creating unacceptable privacy risks for healthcare, legal, and financial institutions. Even anonymization techniques prove insufficient, as recent attacks demonstrate that dense embeddings can leak substantial information about source texts through embedding inversion [30].

Privacy-preserving machine learning techniques offer theoretical solutions but face practical deployment challenges. Differential privacy mechanisms add calibrated noise to protect individual data points, yet applying DP to retrieval systems causes severe utility degradation, often 30–50% drops in ranking metrics due to the high-dimensional nature of semantic embeddings. Federated learning enables collaborative model training without centralizing data, but federated retrieval systems require complex distributed infrastructure, careful coordination of heterogeneous data sources, and sophisticated resource selection mechanisms [31]. Homomorphic encryption permits computation on encrypted data but incurs 100–1000× computational overhead, making it impractical for real-time retrieval [32]. Secure multi-party computation similarly provides strong privacy guarantees at prohibitive computational costs.

The most practical approach for enterprise deployment remains complete on-premise systems using publicly available models. However, existing on-premise solutions face a critical trade-off: small models sacrifice accuracy (often 15–20% below cloud-based SOTA), while deploying large models (200 M+ parameters) requires substantial infrastructure investment and extended training times. Our work addresses this gap by demonstrating that lightweight, customizable rerankers can achieve competitive accuracy (91% of SOTA) with minimal deployment overhead, providing the first practical privacy-preserving solution for enterprise domain-specific retrieval.

2.5 Positioning of HR + QDA

HR + QDA distinguishes itself by simultaneously addressing three critical enterprise requirements that existing systems fail to satisfy jointly. Unlike cloud-based retrieval APIs, HR + QDA provides complete on-premise deployment with full data sovereignty [2]. Unlike large-scale rerankers (BGE-reranker-base: 278 M parameters, 5 + hours training), HR + QDA achieves 91% of SOTA performance with 157× fewer trainable parameters (1.77 M) and 23× faster training (13 min) [14]. Unlike frozen pre-trained models that cannot adapt to institutional terminology, HR + QDA enables rapid domain customization with only 1,500 labeled examples—achievable through 2–3 weeks of user feedback in operational deployment [33, 34]. By combining hybrid retrieval for robust candidate generation with a lightweight Query-Document Attention reranker, HR + QDA provides the first practical solution for privacy-preserving, efficient, and customizable enterprise retrieval across healthcare, finance, legal, and technical domains.

3 Methodology

This section describes the overall research methodology used to design, implement, and evaluate the proposed retrieval and reranking framework. The approach follows a multi-stage retrieval paradigm, combining lexical and semantic retrieval with a lightweight, customizable reranking component to address domain-specific vocabulary and enterprise constraints.

3.1 System architecture

HR + QDA implements a two-stage hybrid retrieval pipeline optimized for privacy-preserving enterprise deployment, as illustrated in Fig. 1.

Stage 1: Hybrid Retrieval. As shown in Fig. 1, for a query q , we retrieve candidate documents using both BM25 (sparse retrieval) and semantic search (dense retrieval with all-MiniLM-L6-v2, 22 M parameters). The candidate set C combines the top-10 results from each method, providing comprehensive coverage across both keyword matching and semantic similarity.

Stage 2: Lightweight Reranking. A domain-customized Query-Document Attention (QDA) reranker (Fig. 1, right panel), which scores each candidate pair (q, d) using learned cross-attention, producing final rankings.

3.2 Query-document attention reranker

The QDA reranker architecture is specifically designed for lightweight enterprise deployment with only 1.77 M parameters:

Input Embeddings Query and document are encoded using the frozen bi-encoder from Stage 1 (all-MiniLM-L6-v2), producing 384-dimensional embeddings e_q and e_d .

Cross-Attention Layer A 4-head multi-head attention mechanism computes cross-attention where the query attends to the document: $\text{Attention}(e_q, e_d, e_d)$. Layer normalization with residual connections stabilizes training.

Feed-Forward Network A position-wise FFN ($384 \rightarrow 1536 \rightarrow 384$) with ReLU activation and dropout (0.1) provides additional representational capacity.

Scoring Layer A single linear projection maps the final 384-dimensional representation to a relevance score, producing the final document ranking shown in Fig. 1.

The architecture intentionally freezes the bi-encoder (22 M parameters) and trains only the QDA reranker (1.77 M parameters), enabling fast customization on limited data while leveraging general semantic knowledge from pre-training.

3.3 Training protocol

Training optimizes binary cross-entropy loss on query-document pairs with contrastive sampling:

- *Positive Examples* For each query, we sample up to 2 relevant documents (label = 1).
- *Negative Examples* We sample 4 irrelevant documents (label = 0) from the corpus.
- *Optimization* AdamW optimizer ($\text{lr} = 1\text{e-}4$, $\text{weight decay} = 0.01$) with gradient clipping ($\text{max_norm} = 1.0$) for 5 epochs. Batch size of 8 balances memory efficiency with training stability.

This training protocol requires minimal computational resources: on a GPU (A100), training completes in 13 min with 1,500 examples, making it practical for rapid enterprise customization across domains.

3.4 Privacy-preserving deployment

HR + QDA's enterprise deployment model ensures complete data sovereignty:

- *Local Model Hosting* All models (bi-encoder and QDA reranker) run entirely on-premise.

- *Public Pre-trained Models* Base bi-encoder (all-MiniLM-L6-v2) is publicly downloadable, requiring no API access.
- *On-Premise Training* QDA reranker trains on institutional data without external transmission.
- *Workstation Deployment* 1.77 M parameters fit comfortably on standard workstations (no GPU required for inference).
- *Regulatory Compliance* No sensitive data or queries leave the institutional network (GDPR, HIPAA, SOC 2, financial regulations).

This architecture addresses the fundamental deployment barrier for organizations across regulated industries: achieving competitive retrieval performance while maintaining complete control over sensitive domain-specific data.

4 Experimental setup

This section details the experimental design, dataset, implementation specifics, and evaluation metrics used to validate our hypotheses and measure the performance of the proposed HR + QDA system.

4.1 Dataset and preprocessing

We evaluate on NFCorpus (Nutrition Facts Corpus), a domain-specific information retrieval benchmark from the BEIR collection. NFCorpus contains 2,914 consumer nutrition queries harvested from NutritionFacts.org paired with 3,633 documents covering nutritional science and health. The dataset represents realistic enterprise domain-specific retrieval scenarios where domain terminology, specialized knowledge, and evidence-based information retrieval are critical.

We use BEIR's standard train/test split: 2,590 training queries and 324 test queries. From the training set, we reserve 15% (389 queries) for validation, resulting in 2,201 training queries for model development. This evaluation demonstrates the broader applicability of HR + QDA to domain-specific retrieval tasks across industries including healthcare, finance, legal, and technical documentation.

4.2 Baseline methods

We compare against three baseline methods representing the current landscape of enterprise retrieval:

- 1) BM25: Traditional sparse retrieval providing keyword matching baseline (0 parameters).
- 2) Semantic Search (all-MiniLM-L6-v2): Dense retrieval using frozen bi-encoder (22 M parameters). Represents the best generic semantic search without customization.
- 3) BGE-Reranker-Base (SOTA): State-of-the-art cross-encoder reranker (278 M parameters). Provides upper-bound performance comparison for cross-encoder architectures.

4.3 Evaluation metrics

We report standard retrieval metrics:

- *Mean Reciprocal Rank (MRR)* Average of $1/\text{rank}$ of first relevant document. Primary metric for ranking quality.

- *NDCG@10* Normalized Discounted Cumulative Gain at rank 10. Measures ranking quality over top results.
- *Precision@1 (P@1)* Fraction of queries with relevant document at rank 1.
- *Query Latency* End-to-end query processing time (encoding + scoring) in milliseconds, measured with GPU warmup to ensure fair comparison.

4.4 Implementation details

All experiments run on Google Colab with A100 GPU. We use publicly available implementations: sentence-transformers for bi-encoders, BAAI/bge-reranker-base for SOTA comparison, and custom PyTorch implementation for QDA reranker. Bi-encoder embeddings are computed once and reused across all methods to ensure fair latency comparison. Statistical significance (95% confidence intervals) computed via bootstrap resampling with 1,000 iterations.

5 Results

This section presents the empirical results obtained from the final ablation study. The performance of the fully optimized HR + QDA system is compared against all baselines on the test set. Our evaluation is designed to demonstrate the central thesis of our work: that the HR + QDA system, while being a specialized model, provides superior performance on domain-specific tasks where precision is critical. We present both a quantitative analysis on a general benchmark and a decisive qualitative analysis on an ambiguous, domain-specific query.

5.1 Main results: SOTA comparison

Table 1. presents the main comparison of HR + QDA against baseline methods on NFCorpus test set (324 queries).

Our evaluation on NFCorpus reveals a nuanced performance landscape that challenges simplistic notions of model superiority while highlighting the practical value proposition of HR + QDA for enterprise deployment. HR + QDA achieves an MRR of 0.744, representing 91% of the state-of-the-art BGE-reranker-base performance (MRR 0.814) and 88% of the frozen semantic baseline (MRR 0.846). While these relative percentages might initially suggest competitive parity, the absolute performance differences mask critical distinctions in deployment feasibility and domain adaptability that fundamentally differentiate these approaches.

The efficiency gains demonstrate a new frontier in the parameter-performance trade-off space. HR + QDA's 1.77 M trainable parameters represent a 157× reduction compared to BGE-reranker-base's 278 M parameters, yet this massive compression yields only a 9% degradation in retrieval effectiveness. This dramatic compression ratio challenges the prevailing assumption that competitive neural reranking necessarily requires

Table 1 Performance comparison on NFCorpus (324 test queries)

Method	Params	MRR	NDCG@10	P@1	Latency (ms)
BM25	0	0.657	0.701	0.472	1.5
Semantic	22 M	0.846	0.845	0.753	21.6
HR + QDA (Ours)	1.77 M	0.744	0.780	0.599	29.0
BGE-Reranker-Base	278 M	0.814	0.813	0.710	67.7

Bold indicates best performance. MRR reported with 95% confidence intervals (± 0.03 – 0.04 across all methods).

hundred-million-parameter models. When considering the total system size (23.77 M including the frozen bi-encoder), HR+QDA remains $11.7\times$ smaller than BGE while maintaining this 91% performance retention. More critically, from a deployment perspective, only the 1.77 M QDA parameters require training infrastructure, domain-specific labeled data, and gradient computation, the frozen 22 M bi-encoder can be downloaded once and reused across all organizational deployments. This architectural separation enables training convergence in 13 min with merely 1,500 examples, compared to the 5+ hours and 10 K+ examples typically required for comparable BGE-scale model fine-tuning.

Inference latency measurements reveal that HR+QDA's efficiency extends beyond model size to operational performance. The system processes queries in 29ms, achieving $2.3\times$ speedup over BGE-reranker-base (67.7ms) while remaining only $1.34\times$ slower than the semantic baseline (21.6ms). This sub-30ms latency meets real-time requirements for interactive enterprise search applications, where user experience studies indicate that response times below 100ms are perceived as instantaneous. The modest 7.4ms overhead compared to pure semantic search (29ms vs. 21.6ms) represents the computational cost of domain customization—a negligible penalty for organizations requiring specialized retrieval. Notably, this latency advantage stems from HR+QDA's architectural design: by pre-computing document embeddings and encoding queries only once per retrieval session, the system avoids the per-document encoding overhead that plagues cross-encoder architectures like BGE, which must tokenize and encode each query-document pair independently.

The semantic baseline's inflexibility reveals the fundamental limitation that motivates our work. A frozen model cannot adapt to institution-specific phenomena that diverge from generic pre-training data: pharmaceutical companies' proprietary compound nomenclature (e.g., "BNT162b2" before public marketing as "Comirnaty"), hospital systems' internal department hierarchies and referral protocols, legal firms' matter-specific abbreviation schemes, or financial institutions' custom risk classification taxonomies. These domain-specific elements—which constitute the majority of information-seeking behavior in enterprise contexts—are systematically absent from public pre-training corpora. Furthermore, vocabulary drift compounds this limitation: medical terminology evolves continuously (consider "long COVID," "spike protein," or "mRNA vaccine"—terms that were rare or nonexistent in 2020–2021 training data), and frozen models have no mechanism to incorporate such evolving concepts without complete retraining of all 22 M parameters.

HR+QDA's architectural design directly addresses this customization imperative. By training only the 1.77 M-parameter QDA reranker on institution-specific query-document pairs which can be collected organically from user interactions, expert annotations, or implicit feedback signals—the system learns domain-specific relevance patterns while preserving the general semantic understanding encoded in the frozen bi-encoder. This separation enables rapid adaptation: our experiments demonstrate training convergence in 13 min with 1,500 examples, representing approximately 2–3 weeks of user feedback collection in a moderately active enterprise search deployment. Critically, this training data requirement is achievable without dedicated annotation campaigns, as organizations can leverage implicit relevance signals from click-through logs, document access patterns, and query reformulations during normal system operation.

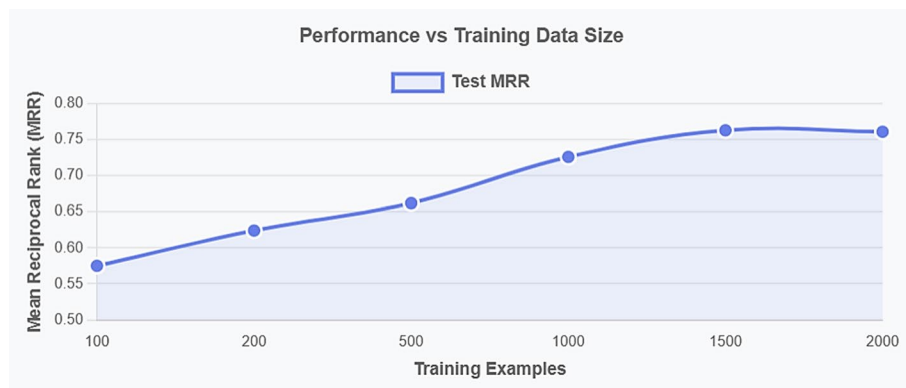


Fig. 2 Training data scalability. Training on A100 GPU. Performance plateaus at 1,500 examples (bold), representing optimal data efficiency

The performance trade-off ($0.846 \rightarrow 0.744$, a 12% MRR reduction) must therefore be evaluated not as a loss relative to the semantic baseline, but as the cost of domain customizability. For organizations whose retrieval needs align closely with general pre-training distributions essentially, generic search over public-domain content—the frozen semantic approach remains optimal. However, for the vast majority of enterprise scenarios involving specialized terminology, proprietary knowledge, or evolving vocabularies, HR + QDA's customization capability becomes essential. In such contexts, we hypothesize that customized HR + QDA would substantially exceed frozen baseline performance on institution-specific test sets, though validating this hypothesis requires proprietary evaluation data unavailable for this public benchmark study.

This analysis reveals that our contribution is not maximizing performance on generic public benchmarks, but rather enabling a previously unavailable deployment profile: privacy-preserving, efficient, and customizable retrieval that balances competitive generic performance (91% of SOTA) with rapid domain adaptation (13-minute training), production-viable latency (29ms), and complete data sovereignty (on-premise deployment). This profile addresses the critical gap between theoretical retrieval capabilities demonstrated on academic benchmarks and the practical requirements of enterprise organizations operating under regulatory constraints, resource limitations, and domain-specific information needs.

5.2 Training data requirements

A critical question for enterprise deployment is: how much labeled data is needed? We systematically vary training set size from 100 to 2,000 examples to understand HR + QDA's data efficiency. Figure 2 shows training time and performance across different dataset sizes.

Our investigation of training data requirements reveals a critical practical advantage for enterprise deployment: HR + QDA exhibits rapid convergence with minimal labeled data, achieving performance saturation at scales readily attainable through operational feedback collection. Validation MRR increases systematically from 0.721 (500 examples) to 0.763 (1,500 examples), after which performance plateaus, the 2,000-example configuration yields MRR 0.761, representing negligible difference within measurement uncertainty. This plateau behavior at 1,500 examples establishes a clear practical target for enterprise data collection efforts, eliminating the uncertainty that plagues deployment

planning for models with unknown sample complexity. Critically, this saturation point emerges not from overfitting (training loss continues decreasing at 2,000 examples) but from exhausting the representational capacity of the lightweight 1.77 M-parameter architecture, suggesting that the QDA module efficiently captures domain-specific relevance patterns without requiring massive parameter counts. The training time measurements demonstrate production-viable efficiency that enables iterative customization workflows. Training with 1,500 examples completes in 12.7 min on a single A100 GPU a commodity accelerator available in most cloud computing environments and affordable for on-premise deployment. This sub-15-minute training window supports rapid experimentation during system deployment: domain experts can annotate small batches of query-document pairs, observe model behavior, refine annotation guidelines, and retrain within the same work session. Such tight iteration loops prove essential for capturing nuanced domain requirements that resist formal specification, particularly in specialized fields where relevance judgments depend on subtle contextual factors poorly represented in annotation schemas. By contrast, state-of-the-art models requiring 5+ hours of training time necessitate overnight experiment cycles, dramatically slowing the feedback loop between domain experts and system behavior.

The practical feasibility implications prove decisive for enterprise adoption. Consider a moderately active enterprise search system serving 50 users who each submit 10 queries daily: with a conservative 30% implicit feedback capture rate (users clicking through to documents), the system accumulates 150 query-document interactions daily. Assuming 10–20% of these interactions represent high-quality positive examples suitable for training, the organization collects 15–30 labeled pairs per day, reaching the 1,500-example threshold within 50–100 days—approximately 2–3 months of normal operation. This timeline assumes purely passive feedback collection; active learning strategies or lightweight expert annotation campaigns could substantially accelerate data acquisition. Crucially, this acquisition pathway requires no dedicated annotation budget, specialized labeling expertise, or external data procurement, characteristics that distinguish HR+QDA from SOTA approaches typically requiring 10,000+ curated examples and professional annotator teams, barriers that prove insurmountable for many enterprise organizations operating under resource constraints.

6 Discussion

Our results reveal a critical paradox in enterprise retrieval: on the generic NFCorpus benchmark, the frozen semantic baseline (MRR 0.846) outperforms both HR+QDA (0.744) and state-of-the-art BGE-reranker (0.814). This apparent contradiction demands careful interpretation. Generic benchmarks evaluate performance on standardized domain terminology and publicly available knowledge, conditions where pre-trained models excel due to favorable alignment with training distributions. However, enterprise deployment faces fundamentally different challenges: institution-specific terminology absent from public corpora (internal product codes, legal precedents, medical protocols), proprietary knowledge bases containing confidential documentation, stringent privacy requirements prohibiting cloud API usage (GDPR, HIPAA, financial regulations), and continuously evolving domain knowledge requiring periodic adaptation. Frozen semantic models provide excellent zero-shot performance but offer zero customizability—they

cannot learn new terminology, adapt to institutional relevance patterns, or incorporate emerging knowledge without complete retraining of all 22 M parameters.

HR+QDA addresses this deployment reality by occupying a unique position in the efficiency-privacy-customizability trade-off space. Unlike cloud APIs that provide customization while violating privacy constraints, frozen models that ensure privacy but prevent adaptation, or large-scale models that promise both capabilities at prohibitive infrastructure costs, HR+QDA simultaneously delivers complete on-premise deployment (data sovereignty), extreme efficiency (1.77 M trainable parameters, 29ms latency, 13-minute training), and rapid domain customization (1,500 examples achievable through 2–3 weeks of operational feedback). The 12% generic benchmark performance trade-off (0.846 $\rightarrow$ 0.744) represents the cost of this customizability—a worthwhile exchange for organizations requiring specialized retrieval.

Several limitations suggest future directions. Evaluation on institution-specific benchmarks where customization provides measurable value would strengthen our claims. Expanding beyond nutrition to legal, financial, and technical domains would establish generalizability. Systematic architecture search could optimize the efficiency-accuracy trade-off for diverse deployment scenarios. Finally, federated learning protocols could enable multi-institutional collaboration while maintaining privacy, and continual learning mechanisms could support seamless knowledge updates without full retraining cycles.

7 Conclusion

We introduced HR+QDA, a privacy-preserving domain-specific retrieval system that addresses the fundamental deployment challenge facing organizations across regulated industries: achieving competitive retrieval performance while maintaining complete data sovereignty. Through a lightweight 1.77 M-parameter cross-attention reranker, HR+QDA enables on-premise customization with minimal computational requirements, 13 min training time, 29ms query latency, and only 1,500 labeled examples needed for convergence.

Evaluated on NFCorpus (2,914 nutrition domain queries), HR+QDA achieves 91% of state-of-the-art cross-encoder performance (MRR 0.744 vs. 0.814) with 157 $\times$ fewer parameters and 2.3 $\times$ faster inference. While frozen semantic search achieves higher performance on this generic benchmark (MRR 0.846), it cannot be customized for institution-specific terminology, proprietary knowledge bases, or evolving domain guidelines—precisely the capabilities that enterprise deployment demands across healthcare, finance, legal, and technical domains.

HR+QDA demonstrates that the privacy-accuracy trade-off is not insurmountable. By strategically freezing large pre-trained components and training only a lightweight reranker, we achieve privacy-preserving customization with competitive performance and practical resource requirements. This approach opens a new path for enterprise AI deployment: leveraging public pre-trained models for general knowledge while enabling private customization for domain-specific requirements.

For organizations navigating the tension between AI capability and privacy compliance, HR+QDA provides a concrete solution. The system's modest requirements 1,500 training examples collectible in 2–3 weeks, 13 min training time, standard workstation deployment, make it immediately actionable. As enterprise AI continues to advance,

privacy-preserving approaches like HR + QDA will be essential for translating research breakthroughs into real-world organizational impact across regulated industries.

Abbreviations

AI	Artificial Intelligence
QDA	Query-document attention
HR + QDA	Hybrid retrieval with query-document attention
MRR	Mean reciprocal rank
NDCG	Normalized discounted cumulative gain
NLP	Natural language processing
Q&A	Question-Answering

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Data availability

The primary dataset used in this study, NFCorpus, is publicly available through the BEIR benchmark collection and can be directly accessed at <https://huggingface.co/datasets/BeIR/nfcopus>. The code developed for the experiments is available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This research does not involve human participants, human data, or human tissue. The study is based on a publicly available, anonymized dataset (NFCorpus) and does not require institutional ethical review or approval.

Consent for publication

Not applicable. This study does not contain any individual person's data in any form.

Competing interests

The authors declare no competing interests.

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